

The updated Utility Tree incorporates insights derived from the ATAM evaluation of the Quality Attribute Scenarios. Performance risks identified under heavy load required the addition of scalability requirements to ensure acceptable latency during peak usage. Interoperability and authentication scenarios revealed strong dependencies on external systems, leading to new data synchronization and external response time ASRs. Concurrent editing and data update scenarios exposed gaps in reliability and consistency, resulting in the introduction of a new quality attribute focused on concurrency control. Maintainability was added to address long-term development sustainability and reduce complexity in external integrations and validation logic.

#	Scenario Summary	Classification	Explanation (≤3 sentences)
1	Search results displayed under heavy load	Risk	Heavy load threatens the <1s latency ASR. No scalability mechanisms are defined in the architecture. User experience and performance may degrade.
2	Account creation stored externally	Tradeoff Point	External storage improves interoperability but introduces performance dependency. Security and latency are affected simultaneously. Architectural decisions must balance speed vs integration.
3	Login loads account externally	Sensitivity Point	Authentication performance depends on external system speed. Small delays strongly impact usability. External latency directly influences system responsiveness.
4	Invalid input validation error message	Non-Risk	Local validation aligns with existing architecture. Minimal complexity and low architectural impact. No conflicts with existing ASRs.
5	Server saves item and confirms to user	Risk	Synchronous server communication may violate latency goals. Reliability and availability requirements are not defined. Performance degradation could occur during peak usage.
6	External systems push updates	Sensitivity Point	Interoperability depends on synchronization timing. Architectural changes in external integration heavily affect consistency. Tight coupling increases system fragility.
7	Concurrent edits on same data	Tradeoff Point	Concurrency control balances performance and data integrity. Current ASRs lack conflict resolution requirements. Design choices affect latency and reliability.

Updated Utility Tree:

